



Brain-To-Brain Communication: Now A Reality

Soldiers able to communicate silently could change the face of warfare on the ground, and perhaps the mind reading technology could prove to be more dangerous than the nuclear bomb

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The brain typically works by sending a signal to peripheral nerves and muscles to induce movement of a limb or any such other action. A brain computer interface (BCI) is a bidirectional communication pathway between the brain and an external device, designed to acquire, analyse, and translate brain signals for a specific action. BCIs provide a new output channel for the brain to communicate with and ultimately control an external device, which could be an artificial limb, a simulated aircraft, or any other device interfaced with a computer.

The human brain cells' messages are transmitted as electrical signals through nerves, and it is still not fully understood as to how the brain codes these electrical signals. Hence it is quite complex reading these signals completely and then transmitting or receiving them surreptitiously over a distance.

Though the ability to send thoughts directly to another person's brain remains somewhat science fiction, technology is now available, in a limited way, to sense thoughts in one's mind and transmit them into another brain.

Experiments have been carried out by planting electrodes inside the human skull to sense the electrical activity produced by the brain cells. Another technology used is non-invasive electroencephalography (EEG)—a monitoring method to record voltage fluctuations representing the activity of the brain.

Decoding received signals has been a problem, hence, respondents have been asked to imagine performing simple tasks (while not actually doing them) so that the electric activity created in the brain could be recorded as EEG patterns. Another challenge was beaming thoughts into someone else's brain. Initially electrodes were used to pass current into the brain to stimulate the imagined thought. To avoid passing current into the brain, a transcranial magnetic stimulation (TMS) wand was used, which induced a current to activate the brain tissue.

In brain-to-brain communication, the 'sender' is asked to concentrate on a task, the electric activity of their brain is recorded using EEG and transmitted to a TMS wand placed on the head of the 'receiver' to stimulate their brain to reproduce the task.

Technology's evolution

Brain-to-brain communication has been of great interest for past many years. Hans Berger, credited with the discovery of the electrical activity of the human brain, recorded it for the first time in 1924 using EEG. In 1969, Fetz showed that monkeys could control the deflection of a biofeedback arm with neural activity.

The research on BCIs began in the 1970s at the University of California while it was in 1977 that Professor Jacques Vidal demonstrated the visually evoked potential controlling a graphical object on a computer screen. In 1988, Bozinovski described controlling multiple start-stop-restart movements of a robot using EEG. In 2000, Nicolelis used BCIs to decode brain activity in monkeys and reproduce their movements in robotic arms. In 2008, photographs published in science journals showed a monkey operating a robotic arm by thinking.

Fig. 1: The MindMesh, an EEG (electroencephalogram) based thinking cap (Source: www.interaction-design.org)

